12.2022

Roll No. ....

CE

UCE–701

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) "Disaster arid development are closely linked". Explain this statement with the help of an example. (CO1)

   (b) Explain briefly the economic, environmental and social impacts of droughts. (CO1)

   (c) What are the different types of hazards ? Explain in brief. (CO1)

2. (a) What are principal driving forces for landslides ? What are the safety tips to be followed during and after landslides ? (CO2)

   (b) What are the consequences of disaster on health services/facilities ? (CO2)

P. T. O.

(c) What are the probable causes of industrial accidents ? What are the methods to be followed for its prevention ? (CO2)

3. (a) Explain the steps like 'early recovery', reconstruction and redevelopment' that are followed during post disaster. (CO3)

(b) What are manmade disasters ? Explain with examples. (CO3)

(c) Explain the term 'incident command system'. Explain briefly the major management roles that are to be fulfilled in ICS organization. (CO3)

4. (a) What is the national policy on disaster management 2009 ? Write down its salient features. (CO4)

(b) What is a nuclear disaster ? What should you do if the nuclear accident had occurred near you ? (CO4)

(c) What lessons are learnt after Uttarakhand tragedy ? Explain the statement "visionless development in sub-Himalayan region". (CO4)

5. (a) Write down the typical causes of floods. Mention some do's and don'ts when someone is stucked in a cyclone. (CO5)

(b) Explain the term 'biological disaster'. Quote an example to explain it and also write down its effect on human population. (CO5)

(c) What is the role of corporate sector in disaster management ? How do they get affected during disaster ? (CO5)

100

Roll No. ....

TCE–712

B. TECH. (CE) (SEVENTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

GEOGRAPHIC INFORMATION SYSTEM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define the GIS and explain the essential components of GIS. (CO1)

   (b) Utility of GIS in the modern era. (CO1)

   (c) How does remote sensing integrate with GIS ? (CO1)

2. (a) What is map projection and transformation ? (CO2)

   (b) "Geographic data consists of spatial data and non-spatial data." Justify the statement. (CO2)

   (c) What are conceptual models in GIS ? (CO2)

3. (a) Give brief about Network Data Structure, Relational Database Structure. (CO3)

P. T. O.

(b) Give brief about Object Oriented Database, Database Management System. (CO3)

(c) "Data quality and sources of errors play a vital role in GIS data." Justify the statement. (CO3)

4. (a) Application of Buffering and Neighbouringhood Functions in GIS. (CO4)

(b) "Interpolation is a process of estimating the value of attributes." Justify the statement and requirement of interpolation. (CO4)

(c) "A network is defined as a set of interconnected linear features." Justify the statement in brief. (CO4)

5. (a) "There are a considerable number of GIS packages available in the market." Justify the statement with a few of the available leading packages. (CO5)

(b) "Web GIS is a GIS system that uses web technologies." Justify the statement, and also explain mobile GIS and Internet GIS. (CO5)

(c) "AM/FM is a subset of GIS." Justify the statement by explaining the components and advantages of AM/FM. (CO5)

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Roll No. ....

TCE–701

B. TECH. (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

DESIGN OF STEEL STRUCTURE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

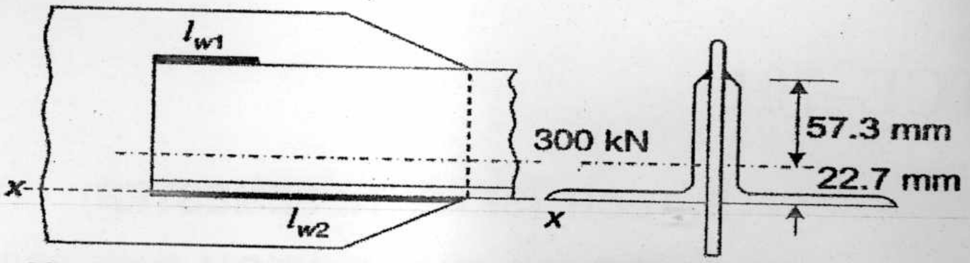
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Two flats (Fe 410 Grade Steel), each 210 mm × 8 mm, are to be jointed using 20 mm diameter, 4.6 grade bolts, to form a lap joint. The joint is supposed to transfer a factored load of 250 kN. Design the joint and determine suitable pitch for the bolts. (CO1)

   (b) A tie member of a truss consists of double angle section, each 80 mm × 80 mm × 8 mm welded on the opposite side of a 12 mm thick gusset plate as shown in figure. Design a fillet weld for making the connection in the workshop. The factored tensile force in the member is 300 kN. (Fe410 grade steel for both connection and members). (CO1)



(c) An ISLC 300 @ 324.7 N/m (Fe 410 grade of steel) is to carry a factored tensile force of 900 kN. The channel section is to be welded at the site to a gusset plate 12 mm thick. Design a fillet weld, if the overlap is limited to 350 mm. (CO1)

2. (a) Design a bridge truss diagonal subjected to a factored tensile load of 300 kN. The length of the diagonal is 3.0 m. The tension member is connected to a gusset plate 16 mm thick with one line of 20 mm diameter bolts of grade 8.8. (CO2)

(b) Compute the tensile strength of an angle section ISA 150 × 115 × 8 mm of Fe 410 grade of steel connected with the gusset plate as shown in figure for the following cases : (CO2)

(i) Gross section yielding

(ii) Net section rupture

(c) With the help of neat diagrams write short notes on the following : (CO2)

(a) Lug Angles

(b) Splices

(c) Gusset plates

3. (a) Design a column section consisting of two channel sections placed back to back for resisting 800 kN of working load. Length of column is 10 metres and is hinged at the two ends. Sections available are ISMC 250, ISMC 300 and ISMC 350. (CO3)

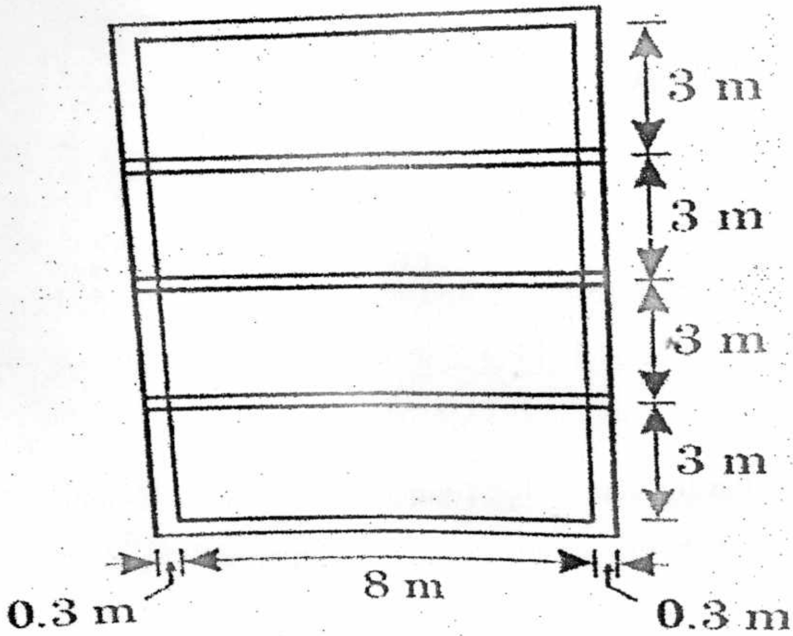
(b) Design the lacing for connecting the two members of lacing and draw the suitable diagram for the column chosen above. (CO3)

(c) An ISA is 100 × 100 × 6 mm is connected to Gusset plate in a truss. The length of strut between the intersection at each end is 4.5 mtrs. Calculate the strength of strut if it is connected by 2 bolts at each end. Treat the end to be fixed-fixed. (CO3)

4. (a) A simply supported steel joist of 4.0 m effective span is laterally supported throughout. It carries a total uniformly distributed load of 40 kN (inclusive of self weight). Design an appropriate section using steel of grade Fe 410. (CO4)

(b) Determine the design bending strength and the design shear strength of a laterally supported beam ISMB 300. (CO4)

(c) A roof of a hall measuring 8 m × 12 m consists of 100 mm thick RCC slab supported on steel I-beams spaced 3 m apart as shown in figure below. The finishing load may be taken as 1.5 kN/m^2. Design the steel beam. (CO4)



5. (a) Design a strut in a roof truss for the following data : (CO5)

Length of the strut 2.235 m

Factored compressive force = 50 kN (due to D. L. and L. L.)

Factored tensile force = 17.80 kN (due to D. L. and W. L.)

Grade of steel Fe 410

Grade of bolts 4.6

Bolt diameter 20 mm

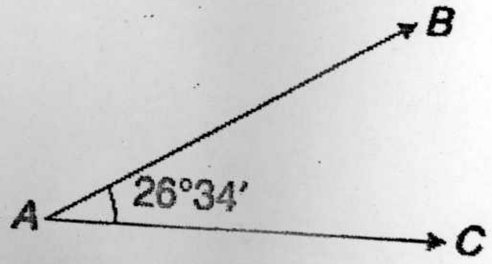
(b) Draw a roof truss and mention various components of roof truss. Draw different types of roof trusses. (CO5)

(c) Design members

AB, AC and joint

A of a roof truss, part of which is shown in for the following data :

Member	Length	Compressive force	Tensile force
AB	2.3 m	75 kN	55 kN
AC	1.8 m	60 kN	80 kN



Design also welded connections at joint A. Use tubes of grade $f_y = 250$ MPA.

(CO5)

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TCE–703

B. TECH. (CE) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

TRANSPORTATION ENGINEERING—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the components of railway tracks with a neat and label diagram. (CO1)

   (b) What are the disadvantages of multi-gauge system ? (CO1)

   (c) Compare railway transport with highway transport. (CO1)

2. (a) What is creep of rails ? Explain the various theories of creep of rails. (CO2)

   (b) What is sleeper ? What are the functions of sleepers ? What are the requirements of good sleepers ? (CO2)

P. T. O.

(c) Explain double headed, bull headed and flat footed rail with the help of neat sketches. (CO2)

3. (a) Explain what is meant by track alignment ? What are the basic requirements of good alignment ? What are the factors which control the alignment of a railway track ? (CO3)

(b) What is the necessity of gradients ? Discuss all types of gradients.(CO3)

(c) If a 6° curve track diverges from the main curve of 4° in an opposite direction in the layout of a B. G. yard, calculate the super elevation and the speed on main line, if the maximum speed permitted on branch line is 45 km. p.h. (CO3)

4. (a) Calculate the lead and radius of a 1 in 8.5 BG turnout for 90 R rails using Coles method. Assume d = 120 mm, α = 6° 42' 35", N = 8.5. (CO4)

(b) What is diamond crossing ? Two BG tracks cross each other at an angle of 1 in 10. Calculate the important dimensions of the diamond crossing. (CO4)

(c) What are the objectives of signalling and interlocking ? (CO4)

5. (a) What are the advantages and disadvantages of railway tunnel ? (CO5)

(b) Give brief notes on the following : (CO5)

(i) Safety precautions in tunneling

(ii) Tunnel ventilation

(c) Describe the different methods of tunneling in hard rocks. (CO5)

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TCE–702

B. TECH. (CIVIL ENGINEERING) (SEVENTH SEMESTER)

END SEMESTER EXAMINATION, Dec., 2022

CONSTRUCTION MANAGEMENT AND PLANNING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Gantt bar Chart. Explain with the help of a suitable example the method of preparing a bar chart. (CO1)

   (b) Explain in detail Risk cost management, and main causes of project failure. (CO1)

   (c) Describe the terms Internal rate of return and Benefit Cost Ratio. (CO1)

2. (a) Illustrate and elaborate project development process. (CO2)

   (b) Illustrate stages and steps involved in project planning. (CO2)

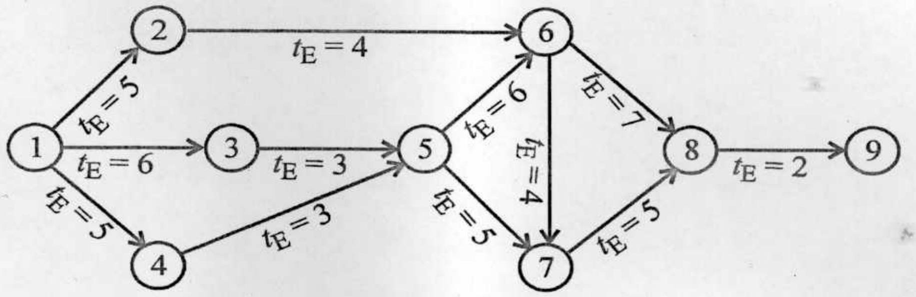
   (c) Illustrate and elaborate categories of construction projects. (CO2)

3. (a) Explain Precedence network with suitable example. (CO3)

P. T. O.

(b) Define 'Latest allowable occurrence time' and derive the expression. (CO3)

(c) Determine the critical path of the network shown below in the figure. Number indicates in the figure indicate time in weeks. (CO3)



4. (a) Explain the term 'Project updating" and its necessity. (CO4)

(b) Explain method of updating a network. (CO4)

(c) Describe cost controlling measures in a construction project. (CO4)

5. (a) Define contract and explain essentials of a contract. (CO5)

(b) Describe the situations for termination of a contract. (CO5)

(c) Describe the safety measures to be adopted in the excavation and in the demolition of a building. (CO5)

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